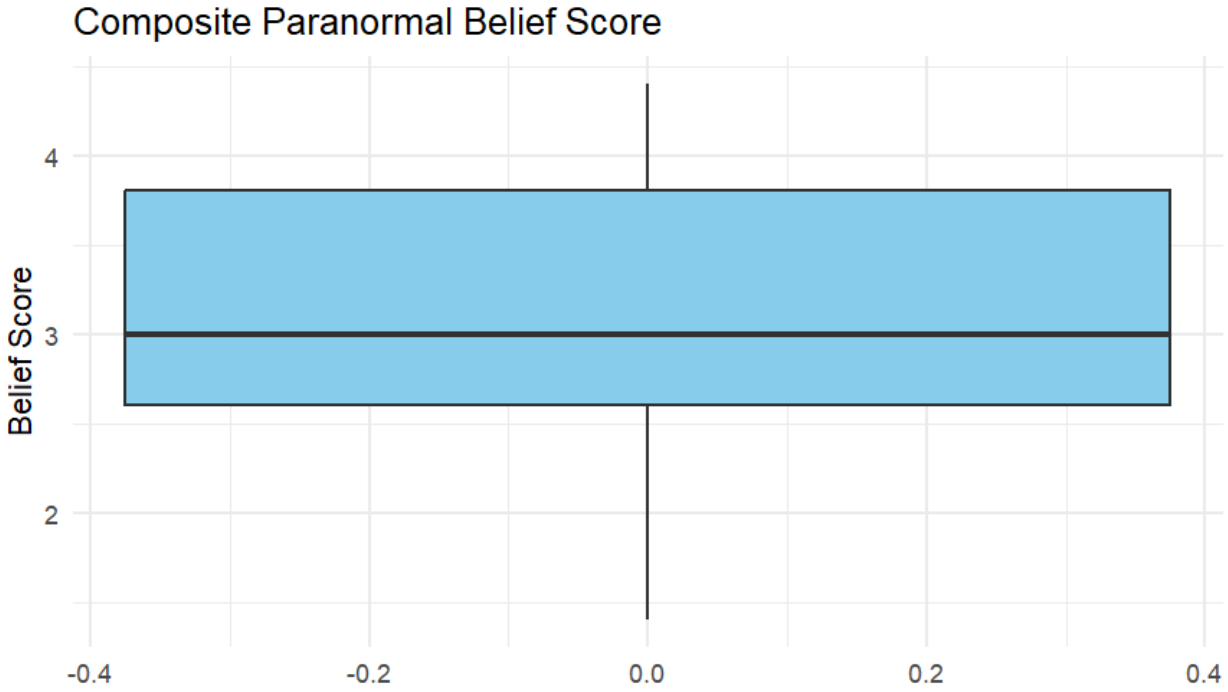
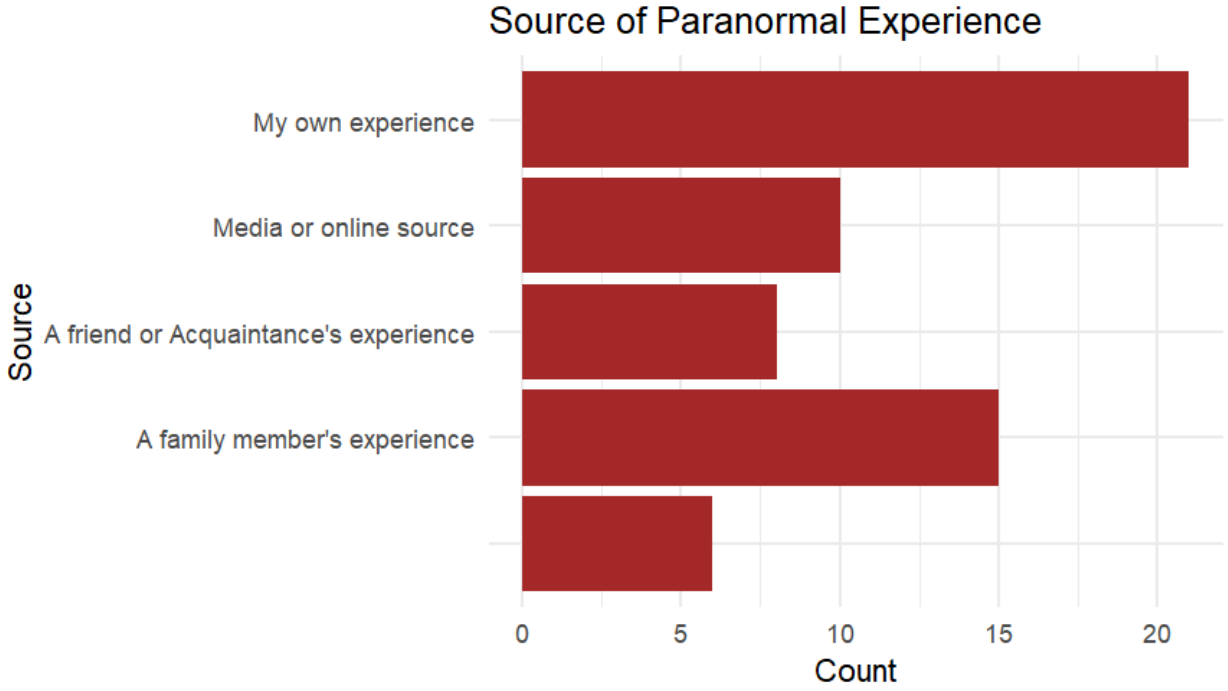
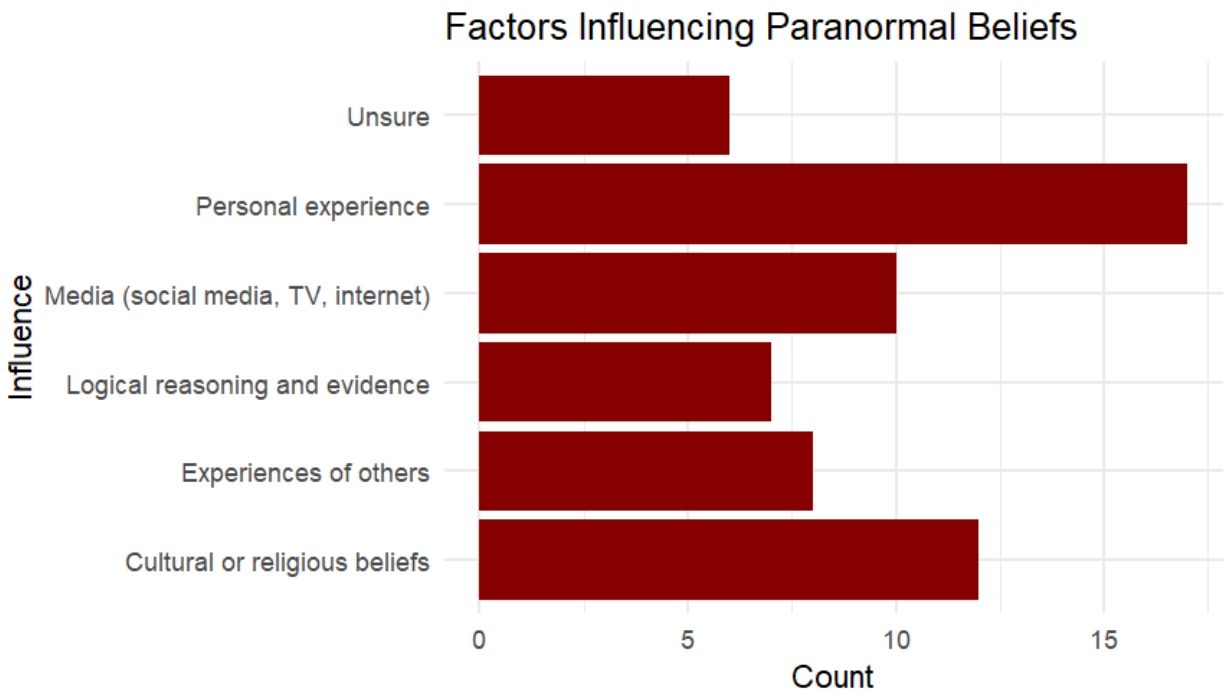
Capstone Project Submission

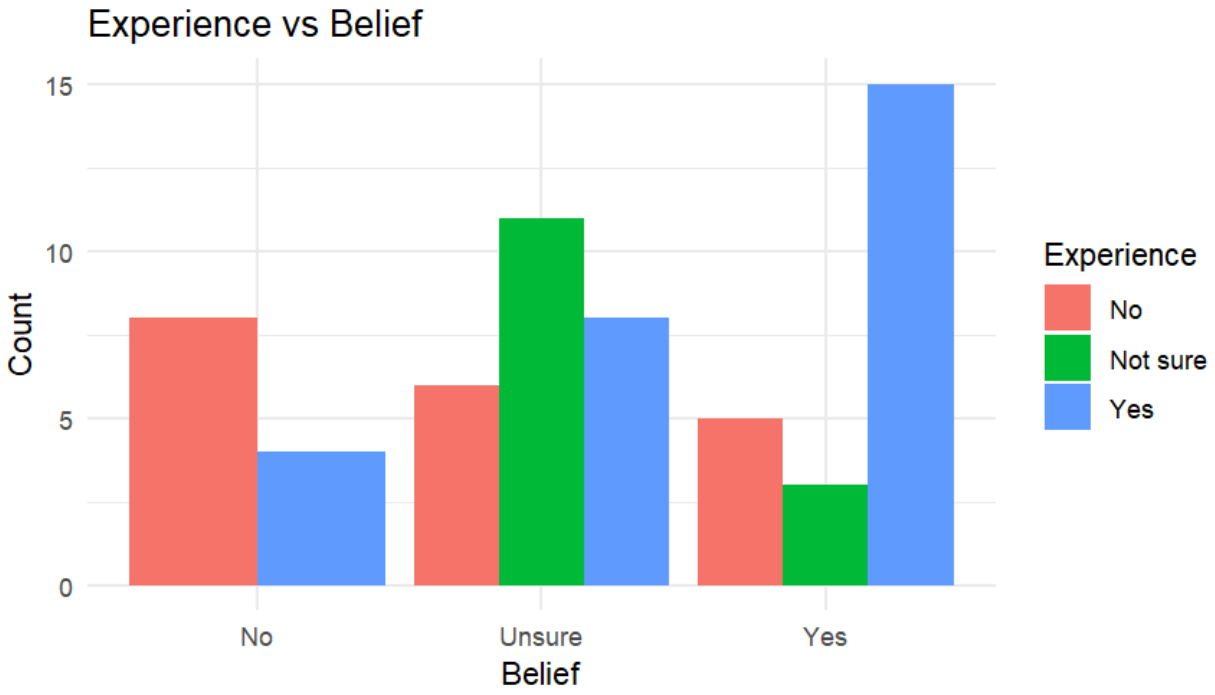
Plots:



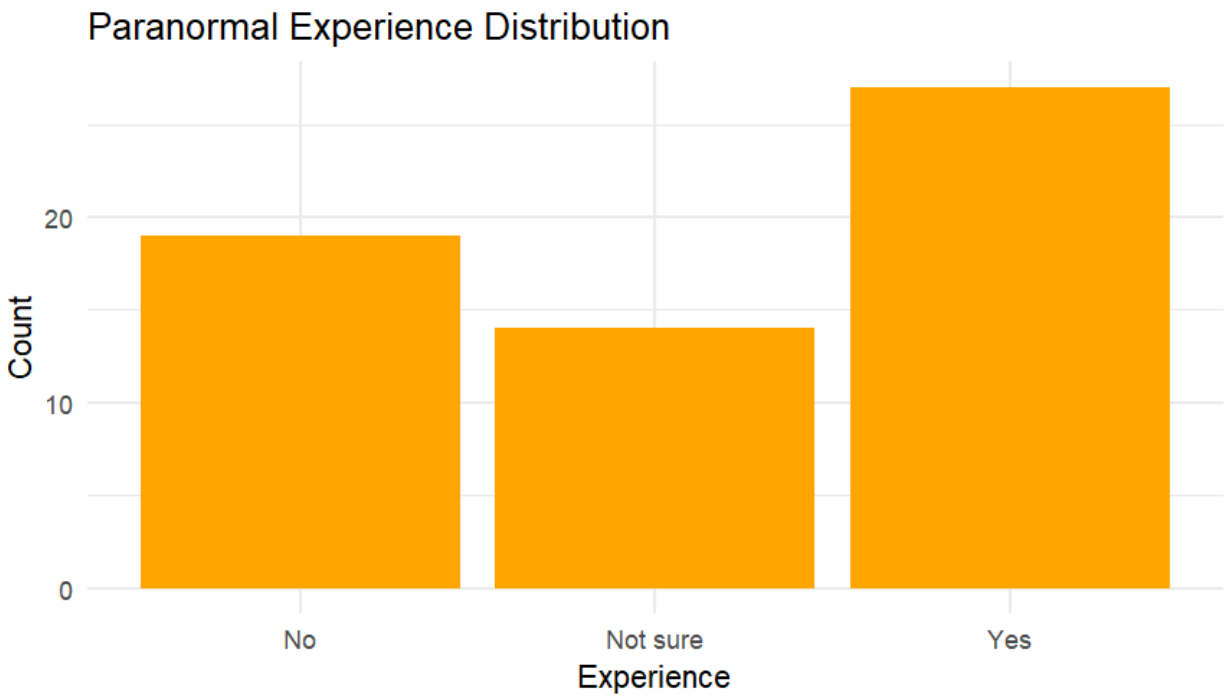


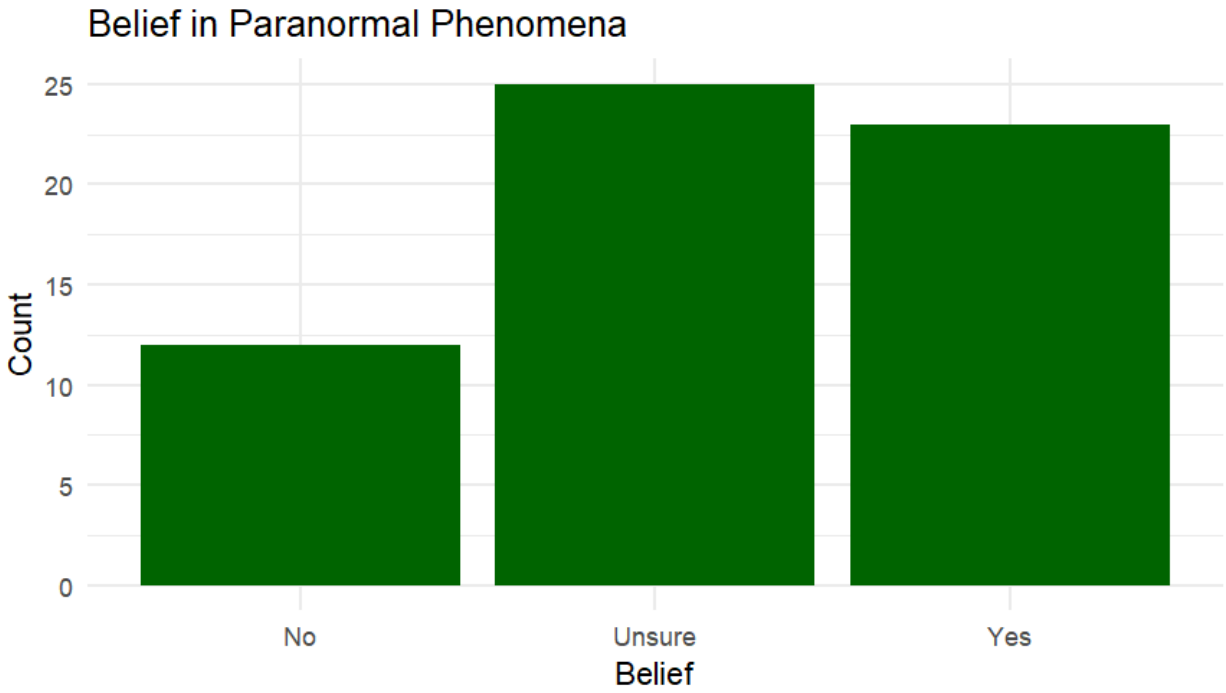
Capstone Project Submission



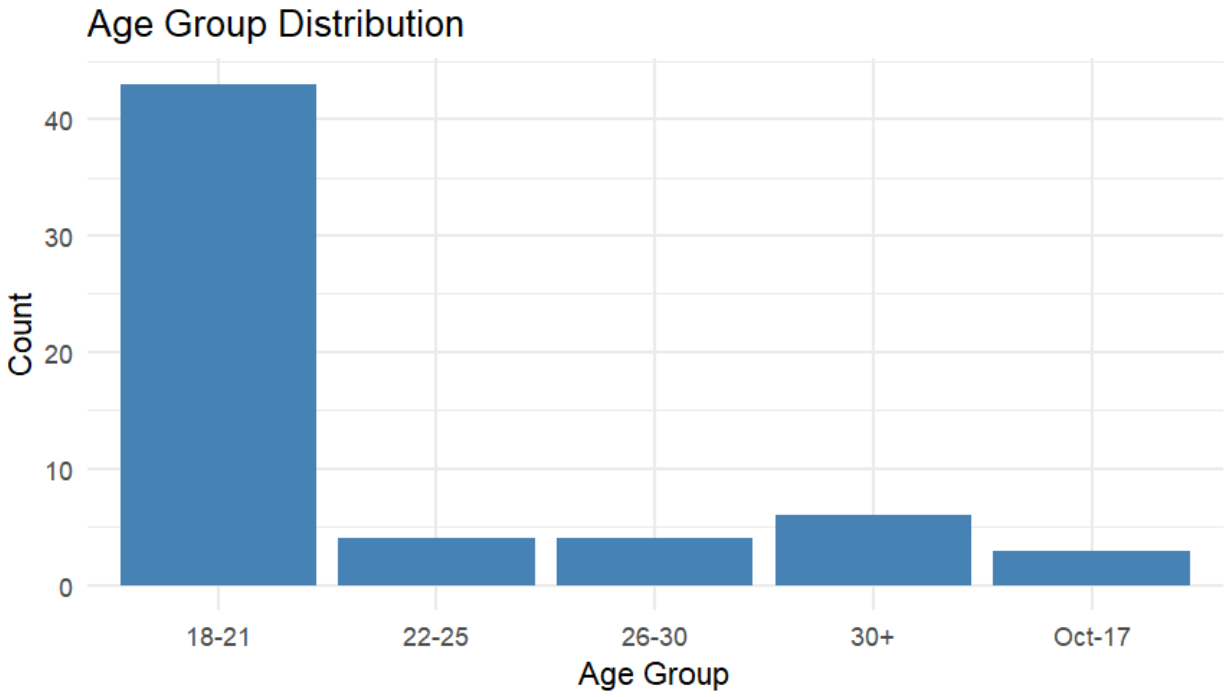


Capstone Project Submission

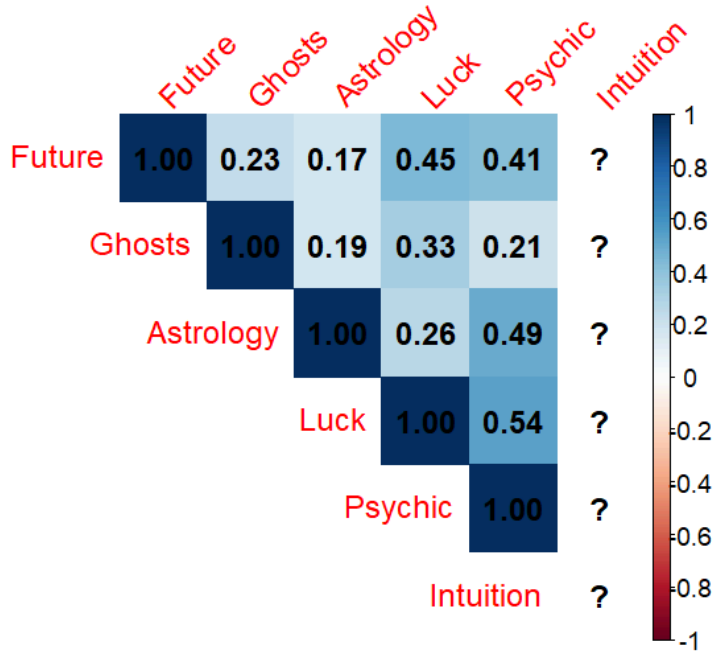


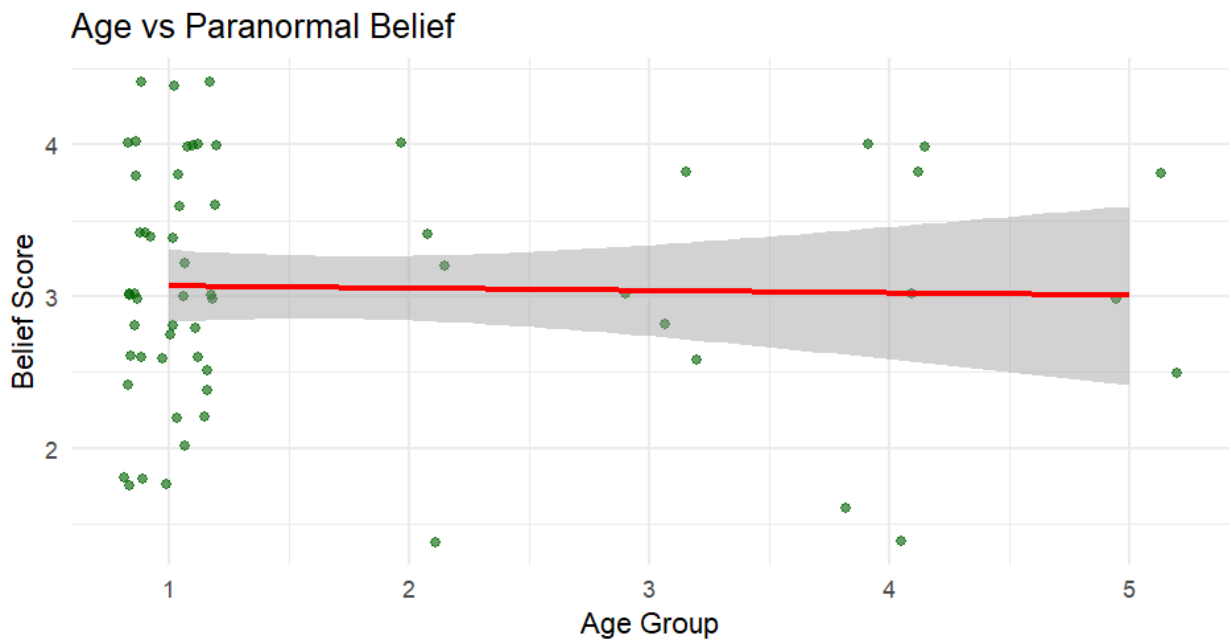


Capstone Project Submission

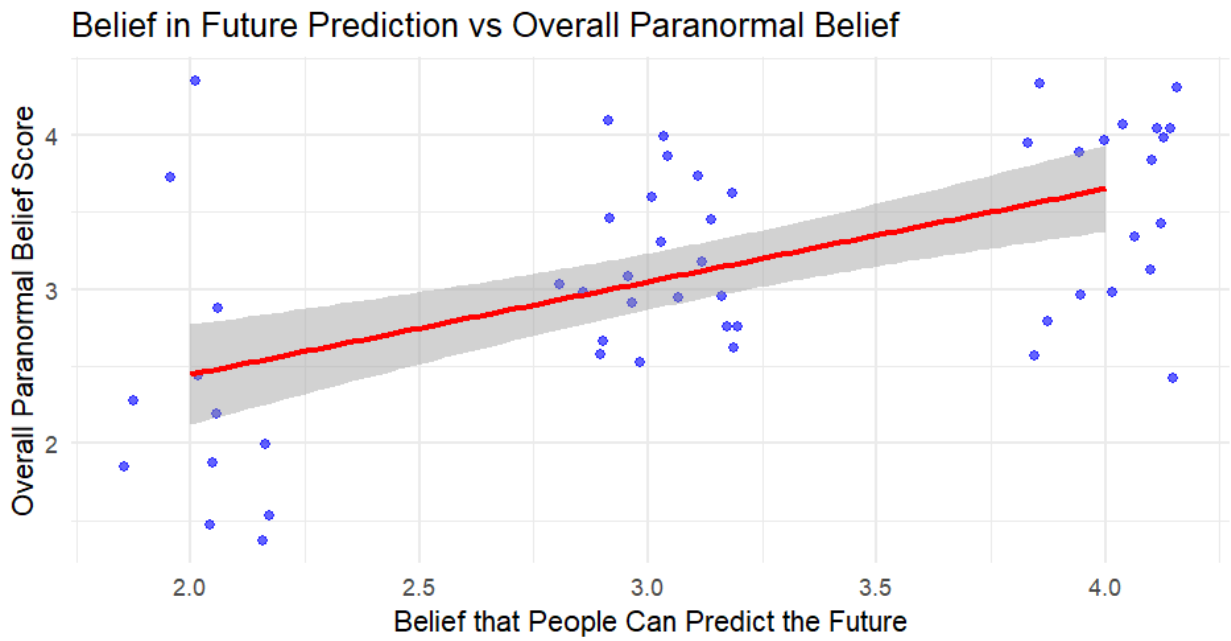


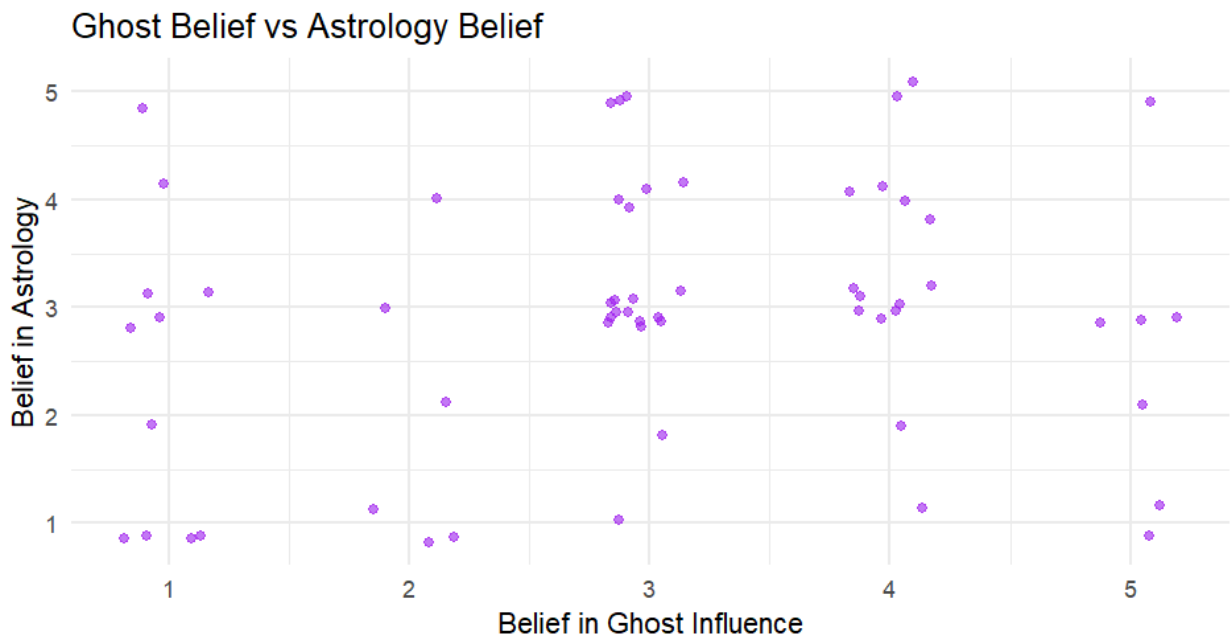
Capstone Project Submission





Capstone Project Submission





Capstone Project Submission

```
> # ====================================================
> # 1 DESCRIPTIVE STATISTICS
> # ====================================================
>
> summary(data$BeliefScore)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.400   2.600   3.000   3.058   3.800   4.400
>
> psych::describe(data$BeliefScore)
   vars  n mean   sd median trimmed  mad min max range  skew kurtosis
X1    1 60 3.06 0.79      3    3.09 0.89 1.4 4.4     3 -0.24     -0.8
    se
X1 0.1
>
```

```
> # ====================================================
> # 2 FREQUENCY TABLES
> # ====================================================
>
> table(data$Gender)

Female   Male
    30     30
>
> table(data$Select.your.Age.Group)

 18-21  22-25  26-30    30+ Oct-17
    43      4      4      6      3
>
> table(data$Do.you.personally.believe.in.paranormal.phenomena.)

    No Unsure    Yes
    12     25     23
>
> table(data$Have.you.ever.experienced.something.you.believe.was.paranormal.)

      No Not sure      Yes
      19       14       27
>
```

Capstone Project Submission

```
> # ====================================================
> # 3 CROSS TABULATIONS
> # ====================================================
>
> table(
+   data$Gender,
+   data$Do.you.personally.believe.in.paranormal.phenomena.
+ )

         No Unsure Yes
  Female  5     11  14
  Male    7     14   9
>
> table(
+   data$Select.your.Age.Group,
+   data$Do.you.personally.believe.in.paranormal.phenomena.
+ )

         No Unsure Yes
  18-21   8     18  17
  22-25   1      3   0
  26-30   0      2   2
  30+     3      1   2
  Oct-17  0      1   2
>
```

```
> # ====================================================
> # 4 ONE SAMPLE T TEST
> # Test if belief score differs from neutral (3)
> # ====================================================
>
> t.test(data$BeliefScore, mu = 3)

        One Sample t-test

data:  data$BeliefScore
t = 0.56175, df = 59, p-value = 0.5764
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 2.852682 3.262318
sample estimates:
mean of x
   3.0575

>
```

Capstone Project Submission

```
>
> # ====================================================
> # 5 INDEPENDENT TWO SAMPLE T TEST
> # Compare belief scores by gender
> # ====================================================
>
> t.test(BeliefScore ~ Gender, data = data)

        Welch Two Sample t-test

data:  BeliefScore by Gender
t = 0.4285, df = 57.404, p-value = 0.6699
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -0.3244063  0.5010729
sample estimates:
mean in group Female   mean in group Male
            3.101667             3.013333
```

```
>
> # ====================================================
> # 6 ONE WAY ANOVA
> # Compare belief across age groups
> # ====================================================
>
> anova_age <- aov(
+   BeliefScore ~ Select.your.Age.Group,
+   data = data
+ )
>
> summary(anova_age)
                      Df Sum Sq Mean Sq F value Pr(>F)
Select.your.Age.Group  4   0.08  0.0198   0.029  0.998
Residuals             55  37.01  0.6729
```

```
> # ====================================================
> # 7 TWO WAY ANOVA
> # Gender + Age effect on belief
> # ====================================================
>
> anova_two <- aov(
+   BeliefScore ~ Gender * Select.your.Age.Group,
+   data = data
+ )
>
> summary(anova_two)
                            Df Sum Sq Mean Sq F value Pr(>F)
Gender                       1   0.12  0.1170   0.171  0.681
Select.your.Age.Group        4   0.03  0.0083   0.012  1.000
Gender:Select.your.Age.Group 2   1.33  0.6650   0.971  0.385
Residuals                   52  35.61  0.6848
>
```

Capstone Project Submission

```
>
> # ====================================================
> # 8 CHI SQUARE TEST
> # Relationship between gender and paranormal belief
> # ====================================================
>
> chisq.test(
+   table(
+     data$Gender,
+     data$Do.you.personally.believe.in.paranormal.phenomena.
+   )
+ )

        Pearson's Chi-squared test

data:  table(data$Gender, data$Do.you.personally.believe.in.paranormal.phenomena.)
X-squared = 1.7803, df = 2, p-value = 0.4106
```

```
> # ====================================================
> # 9 LINEAR REGRESSION
> # Predict overall belief from psychic prediction belief
> # ====================================================
>
> model_linear <- lm(
+   BeliefScore ~ FuturePrediction,
+   data = data
+ )
>
> summary(model_linear)

Call:
lm(formula = BeliefScore ~ FuturePrediction, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.25302 -0.44961 -0.04961  0.35039  1.95381

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)        1.2394     0.3853   3.216  0.00228 **
FuturePrediction   0.6034     0.1203   5.018 6.99e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.651 on 50 degrees of freedom
  (8 observations deleted due to missingness)
Multiple R-squared:  0.3349,    Adjusted R-squared:  0.3216
F-statistic: 25.18 on 1 and 50 DF,  p-value: 6.986e-06

>
```

Capstone Project Submission

```
> # ====================================================
> # 10 LOGISTIC REGRESSION
> # Predict paranormal belief (Yes/No) from belief score
> # ====================================================
>
> # Convert belief variable to binary
> data$BeliefBinary <- ifelse(
+   data$Do.you.personally.believe.in.paranormal.phenomena. == "Yes",
+   1,
+   0
+ )
>
> model_logistic <- glm(
+   BeliefBinary ~ BeliefScore,
+   data = data,
+   family = binomial
+ )
>
> summary(model_logistic)

Call:
glm(formula = BeliefBinary ~ BeliefScore, family = binomial,
    data = data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.9600     1.2358  -2.395   0.0166 *
BeliefScore   0.7971     0.3793   2.102   0.0356 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 79.881  on 59  degrees of freedom
Residual deviance: 74.956  on 58  degrees of freedom
AIC: 78.956

Number of Fisher Scoring iterations: 4
```